

USB4 2.0 ENGINEERING CHANGE NOTICE FORM

Title: Aligning Replenish Timeout parameter usage terminology

Applied to: USB4 Specification Version 2.0

Brief description of the functional changes:

This ECN emphasizes the fact that Replenish Timeout counts missed Handshakes rather than timeout.

Benefits as a result of the changes:

Improves the spec coherency.

An assessment of the impact to the existing revision and systems that currently conform to the USB specification:
--

NA

An analysis of the hardware implications:
--

NA

An analysis of the software implications:
--

NA

An analysis of the compliance testing implications:
--

NA

USB4 2.0 ENGINEERING CHANGE NOTICE FORM

Actual Change

(a). Section 7.3.1.3.2 replenish timeout

To Text:

When an Adapter is in CL1 or CL0s (TX) state, a Downstream Facing Port may “pause” Time Sync Handshakes by not sending a Delay Response to the Upstream Facing Port. The Downstream Facing Port may pause Time Sync Handshakes for maximum of *Replenish Timeout* ~~intervals of~~ *AdapterTimeSyncInterval* ~~skipped handshakes~~ while the Adapter is in CL1 or CL0s (TX) state. The Local Time counter shall continue counting while the Adapter is in CL1 or CL0s (TX) state. If the Adapter stays in CL1 or CL0s (TX) state for more than *Replenish Timeout* ~~times~~ *AdapterTimeSyncInterval* ~~skipped handshakes~~, the Downstream Facing Port will initiate an exit from CL1 or CL0s (TX) state. The Downstream Facing Port shall resume Time Sync Handshakes upon exiting CL1 or CL0s (TX) state keeping the same *AdapterTimeSyncInterval* and the updated Local Time counter value. In the Follow-Up Packet associated with the first Time Sync Handshake after exiting CL1 or CL0s (TX), the Downstream Facing Port shall set the value of the *SleepCyclesN* field to match the number of handshakes missed by the Downstream Facing Port. If the number of missed handshakes is greater than *Replenish Threshold*, after exiting CL1 or CL0s (TX), the Downstream Facing Port will assert an objection to CLx states until it completes *ReplenishN* number of handshakes. See Section 4.2.1.6.3 for more information about objections.

(a). Section 8.2.2.2 TMU Adapter Configuration Capability

To Text:

Table 8-11. TMU Adapter Configuration Capability Fields

DW	Register Name	Bit(s)	Field Name and Description	Type	Default Value
8	TMU_ADP_CS_8	14:0	Replenish Timeout This field contains the maximum of <i>Replenish Timeout</i> skipped handshakes while the Adapter is in CL1 or CL0s (TX) state. <i>AdapterTimeSyncInterval</i> rate for which the TMU will not object to CL1. If an Adapter is in CL1 or CL0s (TX) state for more than this number of skipped time handshakes (<i>AdapterTimeSyncInterval</i> * <i>Replenish Timeout</i>), the TMU shall assert an objection to CL1 state.	R/W	0
		15	Enable Enhanced Uni-Directional Mode (EUDM) When this bit is set to 1b, the USB4 Port shall use Enhanced Uni-Directional Time Sync Handshakes.	R/W	0
		25:16	Replenish Threshold This field contains the number of missed Time Sync Handshakes that requires an objection to CL1 state. If more than <i>Replenish Threshold</i> handshakes are missed, the TMU shall assert an objection to CLx until it finishes at least <i>ReplenishN</i> handshakes. Shall be set to a lower value than in the <i>Replenish Timeout</i> field.	R/W	0
		31:26	Reserved	RO	0